

DDR Timing Parameters

Complete Glossary & Reference Guide

DDR4 (JESD79-4) • DDR5 (JESD79-5)

Memory Controller Capstone Project

Covers all timing symbols, full forms, formulae, constraints, and inter-parameter relationships

Standard	JEDEC JESD79-4 (DDR4), JESD79-5 (DDR5)
Scope	MC Core Timing Reference
BL	BL8 (DDR4), BL16 (DDR5)
Corners	@200 MHz (tCK=5 ns) and @100 MHz (tCK=10 ns)

Contents

Notation and Conventions

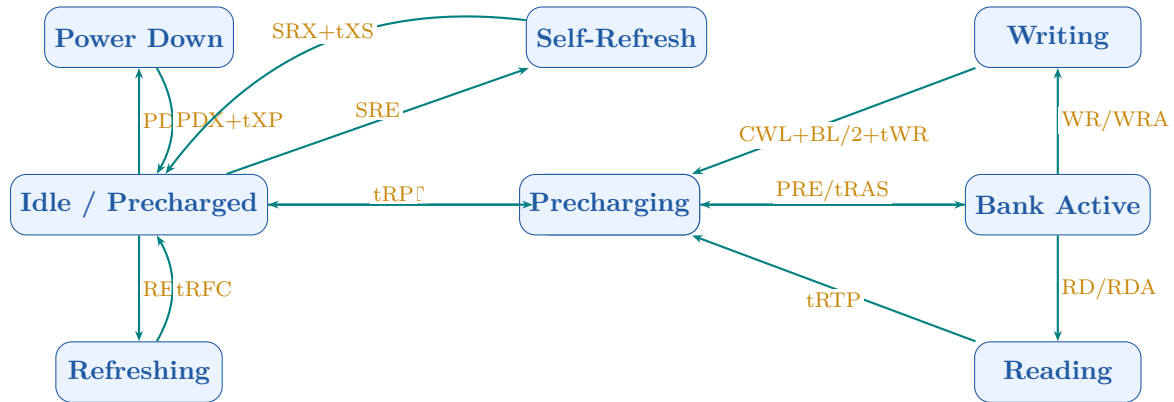
How to read timing values: All timing constraints are expressed as a minimum number of clock cycles (nCK) at the MC input clock. When a spec gives a value in nanoseconds, convert using $nCK = \lceil t_{ns}/t_{CK} \rceil$, subject to any stated minimum nCK floor.

Key Abbreviations

Abbrev.	Full Form	Meaning
nCK	Number of Clocks	Integer clock cycle count at the DRAM/MC interface
ns	Nanosecond	Time unit; $1\text{ ns} = 10^{-9}\text{ s}$
tCK	Clock Period	Reciprocal of clock frequency; $tCK = 5\text{ ns @ } 200\text{ MHz}$
BL	Burst Length	Number of data beats per read/write command. DDR4: BL8, DDR5: BL16
BL/2	Half Burst Length	Number of clock cycles the data bus is occupied. DDR4: 4, DDR5: 8
CL	CAS Latency	Read latency from READ command to first data
CWL	CAS Write Latency	Write latency from WRITE command to first data-in on DQ
AL	Additive Latency	Additional pipeline latency added to CL/CWL (usually 0)
RL	Read Latency	Total read latency = $CL + AL$
WL	Write Latency	Total write latency = $CWL + AL$
BG	Bank Group	Logical grouping of banks; DDR4: $4\text{ BGs} \times 4\text{ banks}$; DDR5: $2\text{ BGs} \times 4\text{ banks}$
AP	Auto Precharge	Automatic precharge at end of read/write burst
PDA	Per-DRAM Addressability	DDR4/5 feature to address individual DRAM in a rank
DFI	DRAM/PHY Interface	Standardised interface between MC and PHY
PHY	Physical Layer	Analogue I/O interface between chip and DRAM
DQ	Data Bit	Single data pin
DQS	Data Strobe	Differential strobe used for data capture
ODT	On-Die Termination	Internal termination resistor in the DRAM
ZQ	Impedance Calibration	Periodic calibration of output driver and ODT impedance
MRS/MRW	Mode Register Set/Write	Command to program DRAM mode registers
MPR	Multi-Purpose Register	DDR4 special register for training patterns
MPC	Multi-Purpose Command	DDR5 replacement for DDR4 MPC/ZQ mechanism
RFM	Refresh Management	DDR5-only rowhammer mitigation; tracks row activations
RAA	Row Activation Accumulator	DDR5 per-bank counter for RFM
ECS	Error Check and Scrub	DDR5 inline ECC scrubbing mechanism
FGR	Fine Granularity Refresh	Configurable tRFC modes (1x/2x/4x)

DRAM States

A DRAM bank cycles through well-defined states. The memory controller must track every bank's current state and issue commands only when the state and all timing constraints are simultaneously satisfied.



Controller responsibility: The MC must maintain per-bank state and timer information for all 16 banks (DDR4: 4 BG × 4) / (DDR5: 2 BG × 4 sub-banks × 2 BGs = 16 banks).

Activation and Precharge Timing

These parameters govern how long a row stays open and the overhead of opening/closing it.

Symbol	Full Form	Typical Value	Explanation
tRCD	RAS-to-CAS Delay	DDR4: 12.5 ns (4 nCK @200) DDR5: 18 ns (4 nCK @200)	Minimum time from ACT command to first READ or WRITE command on the same bank. Represents the time for the sense amplifiers to latch the row data after activation.
tRP	Row Precharge Time	DDR4: 12.5 ns (4 nCK) DDR5: 18 ns (4 nCK)	Minimum time from PRE command until the bank is ready to be activated again. The bitlines must be equalised and the row must return to idle before a new ACT.
tRAS	Row Active Time (min)	DDR4: 35 ns (7 nCK) DDR5: 32 ns (7 nCK)	Minimum time a row must stay activated after ACT before a PRE command may be issued. This ensures the sense amplifiers have fully restored the cell charge. $t_{RAS}(\min) \leq t_{ACT} \leq t_{RAS}(\max)$.

Symbol	Full Form	Typical Value	Explanation
tRAS (max)	Row Active Time (max)	$9 \times t_{REFI}$	Maximum time a row may remain open without a precharge. If exceeded, refresh cannot be serviced correctly. The MC must precharge within this window.
tRC	Row Cycle Time	DDR4: 47.5 ns DDR5: 50 ns	Minimum time from one ACT to the next ACT on the <i>same</i> bank. $t_{RC} = t_{RAS} + t_{RP}$. Prevents consecutive activations before bitline recovery.
tRTP	Internal Read-to-Precharge	$\max(4 nCK, 7.5 ns)$	Minimum delay from a READ command to a PRE on the same bank. The internal sense-amplifier must complete the column read before precharging can begin. For reads with auto-precharge (RDA), hidden precharge starts at RD+tRTP internally.
tWR	Write Recovery Time	DDR4: 15 ns DDR5: 30 ns	Minimum time from the last write data burst until a PRE command may be issued to the same bank. The write data must be fully restored in the DRAM cells before precharge. $WR\text{-to-PRE} = CWL + BL/2 + t_{WR}$.
tDAL	Write-to-ACT Delay (Auto Precharge)	$t_{WR} + t_{RP}$	For write-with-auto-precharge (WRA), the time from the WRA command until a new ACT can be issued to the same bank. $t_{DAL} = t_{WR} + t_{RP}$.

Symbol	Full Form	Typical Value	Explanation
tRRD_L	ACT-to-ACT Delay (Same Bank Group)	max(4 nCK, 4.9–7.5 ns)	Minimum time between two successive ACT commands targeting the <i>same</i> bank group. Higher than tRRD_S because same-BG activations share internal sensing resources.
tRRD_S	ACT-to-ACT Delay (Different Bank Group)	max(4 nCK, 2.5–5 ns)	Minimum time between two successive ACT commands in <i>different</i> bank groups. Relaxed compared to tRRD_L since inter-BG resource conflicts are minimal.
tFAW	Four-Activate Window	DDR4: 35 ns DDR5: 10 ns (tFAW)	A rolling time window in which no more than four ACT commands may be issued across all banks. This limits peak power consumption during multiple simultaneous row activations. The MC must maintain a sliding window counter of the last 4 ACT timestamps.
tPPD	PRE-to-PRE Delay	2 nCK	Minimum time between any two precharge commands (PRE or PREA), regardless of bank. Prevents command bus contention on back-to-back precharge operations.

Read and Write Data Timing

Symbol	Full Form	Typical Value	Explanation
CL	CAS Latency	DDR4: 9–22 nCK DDR5: 22–52 nCK	Number of clock cycles from a READ command to the first data appearing on DQ. Higher clock frequencies require higher CL values (same ns delay).

Symbol	Full Form	Typical Value	Explanation
CWL	CAS Write Latency	DDR4: 9–18 nCK DDR5: CL–2	Number of clock cycles from a WRITE command to the first data-in on DQ. Always $\leq$ CL. In DDR5, CWL = CL–2 per spec Table 9.
AL	Additive Latency	0, CL–1, or CL–2	Extra pipeline latency added before the effective CAS operation. Increases RL/WL without changing the physical CL, useful for command bus scheduling. RL = CL + AL; WL = CWL + AL.
RL	Read Latency	= CL + AL	Total number of clock cycles from a READ command to first valid DQ data. Used in read-to-write turnaround formula: RTW = RL + BL/2 – WL + 2.
WL	Write Latency	= CWL + AL	Total number of clock cycles from a WRITE command to first DQ data driven by the controller. Governs write-to-read turnaround: WTR = WL + BL/2 + tWTR.
tCCD_L	CAS-to-CAS Delay (Same Bank Group, Long)	DDR4: 5–8 nCK DDR5: 8 nCK	Minimum time between two consecutive READ or WRITE commands targeting the <i>same</i> bank group. Longer than tCCD_S because same-BG data path resources (sense amps, I/O gating) are shared.
tCCD_S	CAS-to-CAS Delay (Different Bank Group, Short)	DDR4: 4 nCK DDR5: BL/2 = 8 nCK	Minimum time between two consecutive READ or WRITE commands in <i>different</i> bank groups. In DDR5 this equals BL/2 because the data bus is always occupied for exactly BL/2 cycles.

Symbol	Full Form	Typical Value	Explanation
tWTR_L	Write-to-Read Turnaround (Same Bank Group)	$\max(4 \text{ nCK}, 7.5 \text{ ns})$	After the last write data clock, the minimum number of cycles before a READ command can be issued in the <i>same</i> bank group. Guards DQ bus ownership transfer. Formula: $\text{WTR_cmd} = \text{WL} + \text{BL}/2 + \text{tWTR_L}$.
tWTR_S	Write-to-Read Turnaround (Different Bank Group)	$\max(2 \text{ nCK}, 2.5 \text{ ns})$	Same as tWTR_L but for different bank groups, where bus contention resolves faster. Formula: $\text{WTR_cmd} = \text{WL} + \text{BL}/2 + \text{tWTR_S}$.
tRTW	Read-to-Write Turnaround	DDR4: $\text{RL} + \frac{\text{BL}}{2} - \text{WL} + 2$	Minimum time from a READ command before a WRITE command may follow, ensuring the read DQS postamble does not clash with the write preamble. In DDR4 with 1tCK write preamble: +2; with 2tCK preamble: +3.
tBURST	Burst Duration	BL/2 clock cycles	Time the DQ data bus is occupied for one READ or WRITE operation. DDR4: 4 clock cycles (BL8); DDR5: 8 clock cycles (BL16).
tWPRE	Write Preamble	1 tCK or 2 tCK	The DQS high-Z→low transition before first write data strobe. Default is 1tCK; 2tCK mode is available for higher frequencies (MR4 in DDR4, MR8 in DDR5).
tRPRE	Read Preamble	1 tCK or 2 tCK	The DQS idle-to-active transition window before first read data strobe. 2tCK preamble improves setup margin at higher speeds.

Symbol	Full Form	Typical Value	Explanation
tRPST	Read Postamble	0.5 tCK	Minimum DQS active time after the last read data strobe before DQS returns to high-Z.
tWPST	Write Postamble	0.5 tCK	Minimum DQS active time after the last write data strobe.

Refresh Timing

Refresh is mandatory to prevent data loss. DDR4 and DDR5 differ significantly in refresh granularity and rowhammer management.

Symbol	Full Form	Typical Value	Explanation
tREFI	Average Refresh Interval	DDR4: 7.8 μ s DDR5: 3.9 μ s	Average time between consecutive REFRESH commands at normal temperature ($\leq 85^{\circ}\text{C}$). The MC must issue a REF command at least every tREFI on average. Up to 8 $\times$ postponement is allowed (up to 9 $\times$ tREFI total).
tRFC1	Refresh Cycle Time (Normal mode, 1x)	DDR4/8Gb: 350 ns DDR5/8Gb: 195 ns	Minimum time the DRAM requires after a REFRESH (REFab) command before any other command. All banks are unavailable during this period. Smaller devices have shorter tRFC1.
tRFC2	Refresh Cycle Time (Fine Granularity 2x)	DDR4/8Gb: 260 ns DDR5: N/A	Used in DDR4 Fine Granularity Refresh (FGR) 2x mode where REF is issued at 2 $\times$ normal rate. Each refresh is shorter, reducing latency impact per REF but doubling REF frequency.
tRFC4	Refresh Cycle Time (Fine Granularity 4x)	DDR4/8Gb: 160 ns	Used in DDR4 FGR 4x mode (quadruple refresh rate). Shortest tRFC; used in tXS_FAST path for rapid SR exit.

Symbol	Full Form	Typical Value	Explanation
tRFCsb	Refresh Cycle Time (Same-Bank, DDR5)	DDR5: ~115 ns (8Gb)	DDR5 only. Time required after a same-bank refresh (REFsb) command. Only one bank is refreshed, so all other banks remain accessible, increasing effective bandwidth.
tXS	Exit Self-Refresh to Normal (no DLL lock)	tRFC1 + 10 ns	Minimum time after Self-Refresh Exit (SRX) before any command not requiring a locked DLL. tXS = tRFC1(min) + 10 ns.
tXS_FAST	Exit SR to MRS/ZQCL (DDR4)	tRFC4 + 10 ns	DDR4 only. A fast SRX path allowing Mode Register Set and ZQCL commands before tXS expires, using the shorter tRFC4 base.
tXS_ABORT	Exit SR-Abort to Commands	tRFC4 + 10 ns	DDR4 only. When self-refresh is aborted (CKE raised while SR entry in progress), a shorter wait applies before commands are legal.
tXSDLL	Exit SR to Commands Requiring DLL Lock	tDLLK	After SRX, commands like READ that need a locked DLL must wait until tDLLK cycles after a DLL Reset MRS before they are reliable.

DDR5-Only Refresh Terms

Symbol	Full Form	Value	Explanation
REFab	All-Bank Refresh	Command	Refreshes all banks simultaneously. Equivalent to DDR4 REF. All 16 banks are unavailable for tRFC1 duration.
REFsb	Same-Bank Refresh	Command	DDR5 only. Refreshes a single bank. Other banks remain accessible. tRFCsb ≪ tRFC1. Enables higher REF throughput.

Symbol	Full Form	Value	Explanation
RFM	Refresh Management Command	Command	DDR5 only. Issued when per-bank RAA counter reaches RAAIMT threshold. Adds one extra targeted refresh to mitigate rowhammer.
RAA	Row Activation Accumulator	Counter (per bank)	Per-bank counter in DDR5 that increments on each ACT and decrements by RAADec on each REF. When $RAA \leq RAAIMT$ the MC must issue an RFM command.
RAAIMT	RAA Initial Management Threshold	MR configurable	The RAA value at which the MC <i>must</i> issue an RFM. Typically set to half the rowhammer threshold.
RAAMMT	RAA Maximum Management Threshold	MR configurable	The RAA value above which DRAM may disable internal rowhammer management. Always $\geq RAAIMT$.
tRFM	Refresh Management Cycle Time	~ 195 ns (same as tRFC1)	Time the DRAM requires after an RFM command before the next command on that bank.
ECS	Error Check and Scrub	Feature	DDR5 inline ECC: the DRAM scrubs its own ECC during refresh windows. Controlled by MR14/MR15. tECS governs scrub window.

Power Down and Self-Refresh Timing

Symbol	Full Form	Typical Value	Explanation
tXP	Exit Power Down to Command	$\max(4 \text{ nCK}, 6 \text{ ns})$	Minimum time from Power-Down Exit (PDX, i.e., CKE rising) until a valid command may be issued. During power down, ODT and DLL may be in reduced states that need recovery time.
tCKE	CKE Minimum Pulse Width	$\max(3 \text{ nCK}, 5 \text{ ns})$	Minimum time CKE must be held LOW (for power down) or HIGH (in normal operation). Guards against glitches on the CKE pin.

Symbol	Full Form	Typical Value	Explanation
tCKSRE	Valid Clock Requirement after SRE/PDE	$\max(5 \text{ nCK}, 10 \text{ ns})$	After CKE is deasserted for Self-Refresh Entry (SRE) or Power-Down Entry (PDE), the clock must remain stable and valid for at least tCKSRE before being stopped (SRE) or left running (PDE). Ensures proper internal latching of the mode change.
tCKSRX	Valid Clock Requirement before SRX/PDX	$\max(5 \text{ nCK}, 10 \text{ ns})$	Before asserting CKE for Self-Refresh Exit (SRX) or Power-Down Exit (PDX), the clock must be stable for at least tCKSRX. Ensures the DRAM PLL/DLL has a clean reference before exit.
tCKESR	Min CKE Low Width in Self-Refresh	$t_{CKE} + 1 \text{ nCK}$	Minimum time CKE must remain LOW during self-refresh before it can be raised for SRX. $t_{CKESR} = t_{CKE}(\min) + 1 \text{ nCK}$.
tXPR	Exit Reset to First MRS Command	$\max(5 \text{ nCK}, t_{RFC1} + 10 \text{ ns})$	After power-up reset, minimum delay before the first MRS command. This is the same as tXS since the DRAM runs its internal initialisation like a SR exit.
tACTPDEN	ACT to Power-Down Entry	$1 \text{ nCK (DDR4-1600)}$ $2 \text{ nCK (DDR4-2400+)}$	Minimum time between an ACT command and CKE deassert for power-down entry. Ensures ACT is properly latched before CKE falls.
tRDPDEN	Read to Power-Down Entry	$RL + BL/2 + 1$	Minimum delay from READ command until CKE may fall for power-down. Ensures all read data has been driven out before the DRAM powers down its output drivers.

Symbol	Full Form	Typical Value	Explanation
tWRPDEN	Write to Power-Down Entry (BL8/OTF)	$WL + BL/2 + tWR(nCK)$	Minimum delay from WRITE command until CKE may fall. Write data must be fully written to the DRAM array before power down.
tWRAPDEN	Write-with-AP to Power-Down Entry	$WL + BL/2 + WR + 1$	Similar to tWRPDEN but adds 1 extra cycle for the auto-precharge internal operation.
tREFPDEN	REF to Power-Down Entry	$1-2 nCK$	Minimum delay from REFRESH command until power-down may be entered. The REF command must be latched before CKE falls.
tMRSPDEN	MRS to Power-Down Entry	tMOD (min)	After a Mode Register Set, the DRAM must have tMOD before power-down may be entered. Mode register update must complete before the DRAM powers down its logic.
tPD (max)	Maximum Power-Down Duration	$9 \times tREFI$	The DRAM may remain in power-down for at most $9 \times tREFI$ without losing data. The MC must track power-down entry time and schedule exit before this limit.
MPSM	Maximum Power Saving Mode	Mode	DDR4/5. A deeper power state beyond normal power-down. The clock may also stop. Exit requires additional recovery time (tXMP/tXMPDLL).

Mode Register Set Timing

Symbol	Full Form	Typical Value	Explanation
tMRD	Mode Register Set Command Period	DDR4: 8 nCK DDR5: 16 nCK	Minimum time between two consecutive MRS/MRW commands. Allows the DRAM's internal mode register update logic to settle between writes.
tMOD	Mode Register Set Update Delay	DDR4: $\max(24 \text{ nCK}, 15 \text{ ns})$ DDR5: N/A (uses tMRD)	DDR4 only. Minimum delay from the last MRS command to the first <i>non-MRS</i> command. The DRAM needs tMOD to propagate new register settings into the datapath (CL, CWL, etc.). tMRD governs MRS-to-MRS spacing; tMOD governs MRS-to-any-other spacing.
tMOD (DDR5)	Mode Register Update Delay	DDR5: Not separately defined	In DDR5, MRW (Mode Register Write) timing is governed by tMRD only. tMOD-style behaviour is absorbed into the DFI/PHY command scheduling.
tMPRR	MPR Recovery Time	1 nCK	DDR4 only. Minimum time after MPR read/write before the next command.
tWR_MPR	MPR Write Recovery	tMOD(min)	DDR4 only. After an MRS that modifies the MPR register, tMOD must pass before another command (effectively tMOD clears any MPR state).

ZQ Calibration Timing

ZQ calibration compensates for process, voltage, and temperature (PVT) drift in the output driver and ODT impedance. The DRAM calibrates itself using an external precision resistor on the ZQ pin.

Symbol	Full Form	Typical Value	Explanation
tZQinit	ZQ Initial Calibration Time	1024 nCK	Time required for power-up ZQ long calibration (ZQCL). Issued once during initialisation after all MRS commands are loaded. The DRAM is unavailable for any other command during tZQinit.
tZQoper	ZQ Long Calibration Time (Normal)	512 nCK	Time for a periodic ZQCL command during normal operation (e.g., temperature change). Shorter than tZQinit as the initial calibration has already run.
tZQCS	ZQ Short Calibration Time	128 nCK	Time for a ZQCS (short calibration) command, used for periodic fine-tuning. Less accurate than ZQCL but much faster; typically issued every ~128 μ s.
tZQCAL	ZQ Calibration Start Duration (DDR5)	DDR5: 1 μ s (MPC: ZQCAL Start)	DDR5 only. DDR5 uses MPC commands (MPC ZQCAL Start → wait tZQCAL → MPC ZQCAL Latch) instead of ZQCL/ZQCS.
tZQLAT	ZQ Latch Time (DDR5)	DDR5: 30 nCK	DDR5 only. Time after the MPC ZQCAL Latch command before the DRAM can accept other commands. Allows the calibration result to propagate to the output drivers.

DLL and Clock Timing

Symbol	Full Form	Typical Value	Explanation
tDLLK	DLL Lock Time	DDR4: 512–1024 nCK DDR5: 1024 nCK	After a DLL Reset MRS command, the DLL requires tDLLK clock cycles to re-lock. Commands requiring a locked DLL (primarily READ) are illegal until tDLLK expires. The MC must insert NOPs/DES during this window.
tCK(avg)	Average Clock Cycle Time	Reciprocal of data rate / 2	The average period of the input clock over many cycles. Defines the operating frequency and hence all nCK-based timing values.
tCH(avg)	Average Clock High Pulse Width	$0.48\text{--}0.52 \times tCK$	The high pulse of the differential clock (CK_t high, CK_c low) must occupy 48–52% of each clock period for reliable sampling.
tCL(avg)	Average Clock Low Pulse Width	$0.48\text{--}0.52 \times tCK$	Complementary to tCH; the low portion of each clock period.
tJIT(per)	Clock Period Jitter	$\leq \pm 0.1 \text{ UI}$	Cycle-to-cycle variation of the input clock period. Excessive jitter increases DQ setup/hold margin requirements and can violate CL/CWL timing.
tCPDED	Command Pass Disable Entry Delay	4 nCK	DDR4. After asserting MPSM entry, commands are disabled after tCPDED. The MC must not issue commands after this point until MPSM is exited.

Initialisation Timing

Symbol	Full Form	Typical Value	Explanation
tINIT1	RESET_n Hold Time After Power	200 μ s	DDR5. RESET_n must be held LOW for at least 200 μ s with stable power rails before it may be deasserted. During this time the DRAM initialises internal circuits.
tINIT2	CS_n Hold Before RESET_n Deassert	≥ 10 ns	DDR5. CS_n must be pulled LOW for at least 10 ns before RESET_n is deasserted.
tINIT3	CS_n Hold After RESET_n Deassert	4 ms	DDR5. After RESET_n is deasserted, CS_n must remain LOW for at least 4 ms to allow the DRAM's internal power-on reset sequence to complete.
tINIT4	CK Stable + ODT Settle Time	2 μ s	DDR5. After CK becomes stable, wait 2 μ s for ODT circuitry to settle.
tINIT5	Initial NOPs Duration	≥ 3 nCK	DDR5. After power-up prerequisites, at least 3 NOPs must be issued before the first MRS/MPC command in the init sequence.
tXPR	Exit Reset to First Command	$\max(5 \text{ nCK}, \text{tRFC1}+10 \text{ ns})$	DDR4. After CKE is asserted (following reset), wait tXPR before the first MRS. Equivalent to DDR5 post-CKE init settle time.
tPU_STABLE	Power-Up to Clock Stable	≤ 200 ms ramp	VDD ramp from 300 mV to VDD(min) must complete within 200 ms. VPP must ramp simultaneously or earlier than VDD.

Burst Stop (BST) Timing

BST Command: Terminates an ongoing burst early. For BL8 in DDR4, BST truncates to BC4 if issued 1–3 cycles after the read/write command. For DDR5 (BL16), BST can truncate to BL8. BST is *not* valid after RDA or WRA (auto-precharge in progress).

Symbol	Full Form	Typical Value	Explanation
tBST	Burst Stop Command Delay	1 nCK after burst cmd	BST must be issued at least 1 nCK after a READ/WRITE command. After BST, the bus is released after 2 nCK (turnaround settle).
tBST_settle	Bus Settle after BST	2 nCK	After a BST command, 2 clock cycles must pass before the next read or write command. The data bus must be idle and all DQS transitions must complete.

Key Formulae Summary

Derived Timing Expressions

Row Cycle Time:

$$t_{RC} = t_{RAS} + t_{RP}$$

Read to Write Turnaround (1tCK write preamble):

$$RD\text{-}to\text{-}WR = RL + \frac{BL}{2} - WL + 2$$

Write to Read Turnaround:

$$WR\text{-}to\text{-}RD(\text{same BG}) = WL + \frac{BL}{2} + t_{WTR_L}$$

$$WR\text{-}to\text{-}RD(\text{diff BG}) = WL + \frac{BL}{2} + t_{WTR_S}$$

Write to Precharge:

$$WR\text{-}to\text{-}PRE = CWL + \frac{BL}{2} + t_{WR}$$

Write-with-AP to ACT (same bank):

$$WRA\text{-}to\text{-}ACT = CWL + \frac{BL}{2} + t_{WR} + t_{RP}$$

Read-with-AP to ACT (same bank):

$$RDA\text{-}to\text{-}ACT = t_{RTP} + t_{RP}$$

Write to Refresh (must precharge first):

$$WR\text{-}to\text{-}AREF = CWL + \frac{BL}{2} + t_{WR} + t_{RP}$$

SRX to ACT (no DLL required):

$$t_{XS} = t_{RFC1}(\min) + 10 \text{ ns}$$

tREFI Budget (maximum postponement):

$$t_{\max \text{ between two REF}} \leq 9 \times t_{\text{REFI}}$$

DDR5 Write Latency:

$$CWL = CL - 2$$

tCKESR (DDR4):

$$t_{\text{CKESR}} = t_{\text{CKE}}(\min) + 1 \text{ nCK}$$

DDR4 Read PDE (CL=10, BL8):

$$t_{\text{RDPDEN}} = RL + \frac{BL}{2} + 1 = CL + 4 + 1$$

DDR4 Write PDE (BL8):

$$t_{\text{WRPDEN}} = WL + \frac{BL}{2} + \lceil t_{\text{WR}}/t_{\text{CK}} \rceil$$

DDR4 vs DDR5 — Parameter Comparison

Parameter	DDR4 (JESD79-4)	DDR5 (JESD79-5)	Key Difference / Implication
BL (default)	BL8	BL16	DDR5 doubles burst depth; bus occupied 8 vs 4 cycles. RTW and WTR formulae use BL/2=8.
Banks	16 (4BG×4)	16 (2BG×4×2)	Same bank count; DDR5 adds sub-channel architecture.
tREFI	7.8 μs	3.9 μs	DDR5 doubles REF rate; more bandwidth impact per unit time.
tRFC1 (8 Gb)	350 ns	195 ns	DDR5 significantly shorter tRFC1; internal architecture improvements.
tCCD_L	5 nCK (DDR4-1600)	8 nCK	DDR5 tCCD_L = BL/2; same-BG commands separated by full burst.
tCCD_S	4 nCK	8 nCK	DDR5 tCCD_S also = BL/2; diff-BG no faster than same-BG.
tMRD / tMOD	tMRD=8 nCK, tMOD=24 nCK	tMRD=16 nCK	DDR4 has dual MRS timing (tMRD + tMOD); DDR5 uses only tMRD.
CWL	Separate table (9,11,12...)	CL-2	DDR5 ties CWL to CL tightly; simpler scheduler logic.
tWR	15 ns	30 ns	DDR5 tWR doubled; longer write recovery due to higher density cells.

Parameter	DDR4	DDR5	Key Difference
REFsb	Not supported	Supported	DDR5 same-bank refresh allows per-bank refresh; boosts bandwidth.
RFM	Not supported	Mandatory (RAAIMT)	DDR5 mandates rowhammer management in the MC.
ECS	Not supported	Supported	DDR5 inline ECC scrubbing during refresh windows.
ACT command	1 cycle (CA[13:0])	2 cycles (2-cycle CA)	DDR5 ACT spans two rising edges; Init FSM must handle extended CA encoding.
ZQ mechanism	ZQCL / ZQCS commands	MPC ZQCAL Start+Latch	DDR5 uses MPC encapsulated ZQ flow; Init FSM structure differs.

Command Truth Table Reference

This section lists all valid commands and the states in which they are legal.

Command	Symbol	Legal State / Constraints
Activate	ACT	Bank Idle (Precharged). Constraints: tRP satisfied, tRRD_L/S satisfied, tFAW window not full, tRC satisfied on same bank.
Precharge	PRE	Bank Active. Constraints: tRAS(min) satisfied, tRTP satisfied (after RD), CWL+BL/2+tWR satisfied (after WR).
Precharge All	PREA	Any bank Active or Idle. Same constraints as PRE but applied to all active banks simultaneously.
Read	RD	Bank Active. Constraints: tRCD satisfied, tCCD_L/S satisfied from prior RD/WR.
Read-with-AP	RDA	Bank Active. Same as RD; auto-precharge begins internally at column access + tRTP.
Write	WR	Bank Active. Constraints: tRCD satisfied, tCCD_L/S, tRTW (after RD) satisfied.
Write-with-AP	WRA	Bank Active. Same as WR; auto-precharge begins at CWL+BL/2+tWR internally.
Burst Stop	BST	During active RD or WR burst (not after RDA/WRA). Must be 1+ nCK after burst command.
Refresh	REF/REFa	All banks Idle (Precharged). tRP satisfied on all banks. Constraints: tRFC1 after previous REF.
Same-Bank Refresh	REFsb	DDR5 only. Target bank Idle. tRFCsb after previous command on that bank.
Mode Register Set	MRS/MRW	All banks Idle, all tRP/tDAL/tRFC/tZQ satisfied. tMRD between consecutive MRS; tMOD (DDR4) before next non-MRS.
ZQ Calibration	ZQCL/ZQCS/ ZQ	All banks Idle (tRP, tRFC all satisfied). Blocks all commands for tZQoper/tZQCS/tZQCAL.
Power Down Entry	PDE	Any state (ACT PD or Precharge PD). tACTPDEN/tPRPDEN/tRDPDEN/tWRPDEN satisfied.

Command	Symbol	Legal State / Constraints
Power Down Exit	PDX	Power Down active. Min tCKE low time satisfied. tXP after PDX before commands.
Self-Refresh Entry	SRE	All banks Idle, all timings satisfied. tCKSRE stable clock after CKE falls.
Self-Refresh Exit	SRX	Self-Refreshing. tCKSRX stable clock before CKE rises. tXS after SRX before commands.
RFM	RFM	DDR5 only. Target bank Idle. Issued by MC when $RAA \leq RAA_{\text{IMT}}$. tRFM recovery after.
DESELECT / NOP	DES/NOP	Always legal. No timing constraints. Used to fill idle slots.

Timing Parameter Quick-Reference Card

Symbol	Full Form	DDR4 (8Gb)	DDR5 (8Gb)	Unit
tCK (min)	Minimum Clock Period	0.625 (DDR4-3200)	0.3125 (DDR5-6400)	ns
BL	Burst Length (default)	8	16	beats
CL (min)	CAS Latency (min)	9	22	nCK
CWL	CAS Write Latency	9 (min)	CL-2	nCK
tRCD	RAS-to-CAS Delay	12.5	18	ns
tRP	Row Precharge Time	12.5	18	ns
tRAS(min)	Row Active Time (min)	35	32	ns
tRC	Row Cycle Time	47.5	50	ns
tWR	Write Recovery Time	15	30	ns
tRTP	Internal Read-to-PRE	max(4nCK, 7.5ns)		ns/nCK
tCCD_L	CAS-to-CAS (same BG)	5 (DDR4-1600)	8	nCK
tCCD_S	CAS-to-CAS (diff BG)	4	8 (=BL/2)	nCK
tWTR_L	Write-to-Read (same BG)	max(4, 7.5ns)	max(4, 10ns)	nCK/ns
tWTR_S	Write-to-Read (diff BG)	max(2, 2.5ns)	max(2, 2.5ns)	nCK/ns
tRRD_L	ACT-to-ACT (same BG)	max(4, 7.5ns)	max(8, 5ns)	nCK/ns
tRRD_S	ACT-to-ACT (diff BG)	max(4, 5ns)	max(4, 2.5ns)	nCK/ns
tFAW	Four-Activate Window	35 (2KB page)	10	ns
tREFI	Average Refresh Interval	7800	3900	ns
tRFC1	Refresh Cycle Time (1x)	350	195	ns
tRFC2	Refresh Cycle Time (2x)	260	N/A	ns
tRFC4	Refresh Cycle Time (4x)	160	N/A	ns
tRFCsb	Same-Bank RFC	N/A	~115	ns
tXS	SRX to command (no DLL)	tRFC1+10ns	tRFC1+10ns	ns
tXP	PDX to command	max(4nCK, 6ns)	max(5nCK, 7.5ns)	nCK/ns
tCKSRE	Clock stable after SRE	max(5nCK, 10ns)	max(5nCK, 10ns)	nCK/ns
tCKSRX	Clock stable before SRX	max(5nCK, 10ns)	max(5nCK, 10ns)	nCK/ns
tDLLK	DLL Lock Time	512-1024	1024	nCK
tMRD	MRS-to-MRS	8	16	nCK
tMOD	MRS-to-Non-MRS (DDR4 only)	max(24nCK, 15n)	N/A	nCK/ns
tZQinit	ZQ Init Calibration	1024	tZQCAL (1ps)	nCK
tZQCS	ZQ Short Calibration	128	tZQLAT (30nCK)	nCK
tPPD	PRE-to-PRE Delay	2	2	nCK

MC Core Timing Checker Logic

Implementation note: In the Memory Controller, every clock cycle the timing checker evaluates the following conditions before allowing the command scheduler to issue any command. All conditions must be simultaneously true.

For each bank b and proposed command C at time T :

1. **Bank State:** $\text{state}(b) \in \{\text{valid states for } C\}$
2. **Per-bank timers:** all per-bank timers for C have expired (e.g., tRCD elapsed since last ACT, tRAS elapsed since ACT for PRE, etc.)
3. **Cross-bank timers:**
 - tRRD_L: no ACT to same BG within last tRRD_L cycles
 - tRRD_S: no ACT to any BG within last tRRD_S cycles
 - tFAW: fewer than 4 ACTs within the last tFAW cycles
 - tCCD_L/S: appropriate CCD constraint since last RD/WR in same/diff BG
 - tWTR_L/S: appropriate WTR constraint since last write data (after WR burst end)
4. **Refresh budget:** the refresh budget counter has not expired (i.e., $T_{\text{last REF}} + \text{postponement} < 9 \times \text{tREFI}$)
5. **DDR5 only — RAA:** if command is ACT, $\text{RAA}(b) > 0$ (else RFM must be issued first)

Glossary Index

Symbol	Full Form
ACT	Activate command
AL	Additive Latency
AP	Auto Precharge
BG	Bank Group
BL	Burst Length
BST	Burst Stop
CKE	Clock Enable
CL	CAS Latency (Column Address Strobe Latency)
CWL	CAS Write Latency
DES	Deselect command (no-operation with CS_n high)
DFI	DRAM/PHY Interface
DLL	Delay-Locked Loop
DM	Data Mask
DQ	Data Bit
DQS	Data Strobe
ECS	Error Check and Scrub (DDR5)
FGR	Fine Granularity Refresh
MPC	Multi-Purpose Command (DDR5)
MPR	Multi-Purpose Register (DDR4)
MRS	Mode Register Set (DDR4)
MRW	Mode Register Write (DDR5)
NOP	No-Operation command
ODT	On-Die Termination

(continued)

Symbol	Full Form
PDA	Per-DRAM Addressability
PDE	Power Down Entry
PDX	Power Down Exit
PHY	Physical Layer interface
PRE	Precharge (single bank)
PREA	Precharge All (all banks)
RAA	Row Activation Accumulator (DDR5)
RAAIMT	RAA Initial Management Threshold (DDR5)
RAAMMT	RAA Maximum Management Threshold (DDR5)
RD	Read command
RDA	Read with Auto Precharge
REF	Refresh command (DDR4) / REFab All-Bank Refresh (DDR5)
REFsb	Same-Bank Refresh (DDR5 only)
RFM	Refresh Management (DDR5 rowhammer mitigation)
RL	Read Latency (= CL + AL)
SRE	Self-Refresh Entry
SRX	Self-Refresh Exit
tACTPDEN	Timing of ACT to Power-Down Entry
tBURST	Burst Duration (= BL/2 clock cycles)
tCCD_L	CAS-to-CAS Delay, same Bank Group (Long)
tCCD_S	CAS-to-CAS Delay, different Bank Group (Short)
tCK	Clock Cycle Time
tCKE	CKE Minimum Pulse Width
tCKESR	CKE Low Width in Self-Refresh
tCKSRE	Valid Clock after Self-Refresh Entry
tCKSRX	Valid Clock before Self-Refresh Exit
tCPDED	Command Pass Disable Entry Delay
tDAL	Write-to-Activate Delay (Write-with-AP)
tDLLK	DLL Lock Time
tFAW	Four-Activate Window
tINIT1–5	Initialisation timing steps (DDR5)
tMOD	Mode Register Set Update Delay (DDR4)
tMRD	Mode Register Set Command Period
tPD(max)	Maximum Power-Down Duration
tPPD	PRE-to-PRE Delay
tPRPDEN	PRE to Power-Down Entry
tRAA	Row Activation Accumulator timing (DDR5)
tRAS	Row Active Time
tRC	Row Cycle Time (= tRAS + tRP)
tRCD	RAS-to-CAS Delay
tRDPDEN	Read to Power-Down Entry
tREFI	Average Refresh Interval
tRFC1	Refresh Cycle Time, 1x mode
tRFC2	Refresh Cycle Time, 2x mode (DDR4)
tRFC4	Refresh Cycle Time, 4x mode (DDR4)
tRFCsb	Same-Bank Refresh Cycle Time (DDR5)
tRFM	Refresh Management Cycle Time (DDR5)
tRP	Row Precharge Time
tRPRE	Read Preamble
tRPST	Read Postamble

(continued)

Symbol	Full Form
tRRD_L	Activate-to-Activate Delay, same Bank Group
tRRD_S	Activate-to-Activate Delay, different Bank Group
tRTP	Internal Read-to-Precharge Delay
tWPRE	Write Preamble
tWPST	Write Postamble
tWR	Write Recovery Time
tWRAPDEN	Write-with-AP to Power-Down Entry
tWRPDEN	Write to Power-Down Entry
tWTR_L	Write-to-Read Delay, same Bank Group
tWTR_S	Write-to-Read Delay, different Bank Group
tXP	Exit Power-Down to Command
tXPR	Exit Reset to First Command
tXS	Exit Self-Refresh to Command (no DLL)
tXS_ABORT	Exit SR-Abort to Command
tXS_FAST	Exit SR to MRS/ZQCL (DDR4)
tXSDLL	Exit SR to Command Requiring DLL Lock
tZQCAL	ZQ Calibration Duration (DDR5 MPC)
tZQCS	ZQ Short Calibration Time
tZQinit	ZQ Initialisation Calibration Time
tZQLAT	ZQ Latch Time (DDR5)
tZQoper	ZQ Long Calibration Time
WL	Write Latency (= CWL + AL)
WR	Write command
WRA	Write with Auto Precharge
ZQ	Impedance Calibration
ZQCL	ZQ Calibration Long
ZQCS	ZQ Calibration Short